Module: COMP1844	Information Analysis and Visualisation
Coursework	Contribution: 100% of the grade
Module leader: Konstantin Kapinchev	Due date: 22/04/2026
Approximate time to complete the coursework: 50 hours	
Learning outcomes: 1. Identify and discuss fundamental concepts related to information analysis and visualisation 2. Demonstrate an understanding of different types of information visualisation and identify appropriate types of visualisation for various types of data 3. Apply analytical and visualisation tools and techniques to obtain insight from datasets	

Plagiarism is presenting somebody else's work as one's own. It includes copying information directly from online resources or books without referencing the material, submitting joint coursework as an individual effort; copying another student's coursework, purchasing coursework from someone else and submitting it as own work. Suspected plagiarism will be investigated and if found to have occurred will be dealt with according to the procedures set by the University. All material copied or amended from any source must be referenced correctly according to specific reference style.

Courseworks will be submitted for electronic plagiarism checks.

Coursework Submission Requirements:

- An electronic copy of the coursework is expected to be uploaded on Moodle by the deadline

- The last uploaded version will be the one that is marked

- The limit of the file size is 100 MB

- Grade and feedback will be available on Moodle

- The coursework cannot be submitted via email or as a paper copy

- In order to be marked, the source code needs to meet the following requirements:
    1. **To be error free**
    2. **To generate the required visualisation and results**

- The submitted files are expected to be virus-free, not protected by a password or corrupted

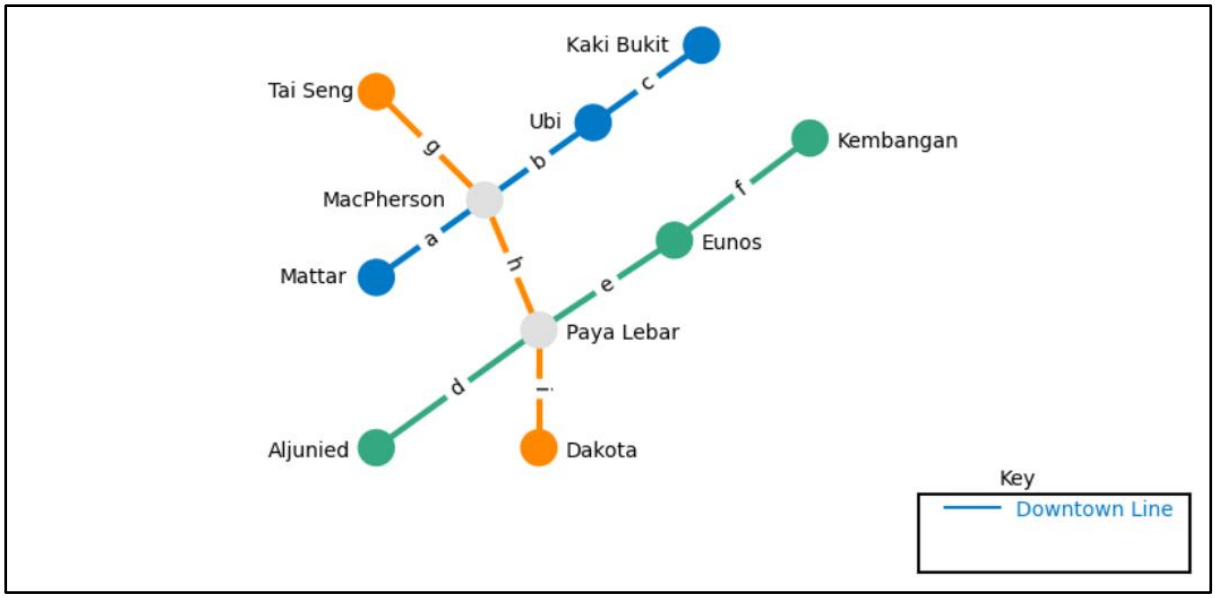
The University website has details of the current Coursework Regulations, including details of penalties for late submission, procedures for Extenuating Circumstances, and penalties for Assessment Offences. See https://www.gre.ac.uk/student-services/regulations-and-policies for details.

Use of AI

In case AI-generated content is included in the coursework, it needs to be enclosed in quotation marks and properly referenced. This applies to both text and source code. Otherwise, if detected, it may be considered plagiarism.

Coursework Specification:

Consider the following image, generated by a **Python program**, which represents part of the public transport network of Singapore:



Task 1

By using **graph data structure**, develop a **Python program**, which generates the above image. Add an **edge attribute**, which represents the distances between stations in kilometers. Use online map services, such as Apple Maps or Google Maps, to approximate the distances. Replace the edge labels, which are currently letters, with the values from the edge attribute.

In its current state, the map shows only three lines and ten stations. Expand the **Python program** by adding more **lines** and more **stations**. Follow a consistent colour scheme. The resulting map is expected to have a **minimum of five lines** with a **minimum of five stations** on each line, including interchange stations. The generated graph is expected to be **connected**, that is, there is always at least one route between any two stations. The generated graph is expected to represent accurately a section of the actual map of the transport network. **A complete map of the transport network is not expected**. Complete the **Key** according to the generated map. By following exactly the same style as shown in the image above, add distances in kilometers between all stations. Add a second **edge attribute**, which represents the distances in miles between the stations.

At the start of the program, allow the user to choose which edge attribute to visualise on the graph, the attribute representing distances in **kilometers** or **miles**.

Task 2

Extract the following data from the graph generated in **Task 1**:

Total length of the transport network in kilometers and miles

The average distance between stations connected with single edge in kilometers and miles

The coursework is expected to have the following documents:

1. Python source code file implementing Task 1 and Task 2 with file extension "py"

2. PDF document with the following content:

1. **Image generated by source code for Task 1**
2. **Data generated by source code for Task 2**

In order to be marked, the Python program is expected to use only the following four Python libraries:

1. **NumPy**
2. **pandas**
3. **NetworkX**
4. **Matplotlib**

While it is not necessary for all suggested libraries to be used, no other libraries are expected to be imported.

The following are the expected first four lines of the Python program:

```
import numpy as np
import pandas as pd
import networkx as nx
import matplotlib.pyplot as plt
```

Marking Scheme

Task	Achieved well	Partly achieved	Poorly achieved or not achieved
Task 1			
Transport Network (60%)			
Visualised graph consists of minimum 5 lines with minimum 5 stations on each line	10-8	7-3	2-0
The graph is connected	5-4	3-2	1-0
Consistent colour scheme is utilised	5-4	3-2	1-0
The generated graph represents accurately section of an actual map of the transport network	10-8	7-3	2-0
Accurate Key is visualised	10-8	7-3	2-0
Edge labels are utilised (kilometers and miles)	10-8	7-3	2-0
User can select which edge label to visualise	10-8	7-3	2-0
Quality of the Visual Representation (30%)			
Graphical elements do no cluster or overlap	10-8	7-3	2-0
No large unused plotting spaces	10-8	7-3	2-0
No visual clutter	10-8	7-3	2-0
Task 2 (10%)			
Total length in kilometers and miles	5-4	3-2	1-0
Average distance in kilometers and miles	5-4	3-2	1-0